Infrastructure, Borders, and Conflict in East Africa

Project Report

GISC412

Advanced Methods in Geospatial Data Science

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1. INTRODUCTION

Violent conflict remains one of the most notable challenges in the 21st century. While geopolitical discourse primarily focusses on conflict in the Middle East, Sub-Saharan Africa continues to experience significant levels of conflict, with around 30 percent of countries affected (Fang et al., 2020), carrying severe consequences for large populations. Understanding the spatial drivers of conflict is important for policymakers, humanitarian organisations, and any other parties involved in designing interventions or resource allocation.

Traditionally, it has been suggested that a strong state presence and provision of public services should reduce conflict due to an authoritative presence and addressed grievances (Besley & Persson, 2009, 2010), but in reality, the drivers of conflict are more nuanced. An authoritative presence can exacerbate tensions and uneven service provisions can itself be a driver of conflict (Besley & Persson, 2009). Four countries in East Africa – Kenya, South Sudan, Uganda, and Ethiopia – while geographically neighbours, have strong diversity in infrastructure, economic activity, and resource scarcity. This makes them good candidates for investigating spatial drivers of conflict.

This study employs geographic weighted regression to analyse the relationship between state infrastructure and armed conflict across these four countries, utilising Open Street Maps (OSM) data for infrastructure and the Armed Conflict Location and Event Data Project (ACLED) dataset from 2015 to 2024. This ten-year period captures significant political transitions including South Sudan's ongoing civil war (Global Centre for the Responsibility to Protect, 2025), Ethiopia's recent conflicts in Tigray, Oromia, and Amhara regions (Council on Foreign Relations, 2024), Kenya's contested election (Wilson Center, 2022), and Uganda's evolving security landscape. This study seeks to examine if different state services reduce violence or if effects are context-dependent and poses the following question:

To what extent do the relationships between state infrastructure (police presence, healthcare facilities, and educational institutions) and conflict events vary spatially across East Africa, and what does this spatial heterogeneity reveal about the context-dependent nature of state capacity in conflict dynamics?

2. DATA AND METHODS

The data was obtained from multiple sources.

- Conflict Data: The Armed Conflict Location & Event Data (ACLED) dataset provided detailed records of political violence, including event location (latitude/longitude), date, event type, actors, and estimated fatalities. Obtained from https://acleddata.com.
- Population Data: WorldPop (2020) population raster data (~100 m resolution) was used to normalize other data into per-capita rates. Obtained from https://www.worldpop.org.
- Infrastructure Data: Police, Hospitals and Schools vector data was obtained from OpenStreetMap (OSM) programmatically using OSMnx.
- Borders and Administrative Boundaries: GADM polygons were used to define country and admin-2 boundaries. Obtained from https://gadm.org/download_country.html.

The data was loaded using a dedicated module. The ACLED data was read in and filtered to only include events from 2015-2024. For each country the population raster and boundaries were read. Each OSM feature was loaded in, either through the API, or accessing a saved parquet file if the data had already been downloaded in a previous run. Each dataset was reprojected to EPSG:32637 and rows with empty geometries were dropped. The population rasters used -99999 to denote a cell with no data and these were replaced with 0.

A dedicated feature engineering module was created. The ACLED data and each of the OSM features were spatially joined to the admin units and the count of features per unit were recorded. The CRS alignment of the boundaries and the population rasters were confirmed then the populations were summed for each admin unit. Each of the features were checked for CRS alignment and then features were calculated per capita, using per 10k for OSM features and per 100k for ACLED data. A log transform of each feature was created to be used for modelling.

A modelling module was also created. An OLS regression was run as a baseline for comparison. The GWR used the mgwr module. An optimal bandwidth was selected using mgwr.sel_bw and the GWR was fitted using the log features with the conflict data as the target.

Each of these modules was called in turn from the project notebook and the resulting coefficients and model R-squared were plotted for analysis. The feature engineering and modelling was run once per country.

3. RESULTS AND DISCUSSION

The police station density coefficient was plotted first and is presented as Figure 1. It is important to note that the models were run on each individual country, the coefficients are in the same scale, but spatial correlation is only modelled within each country.

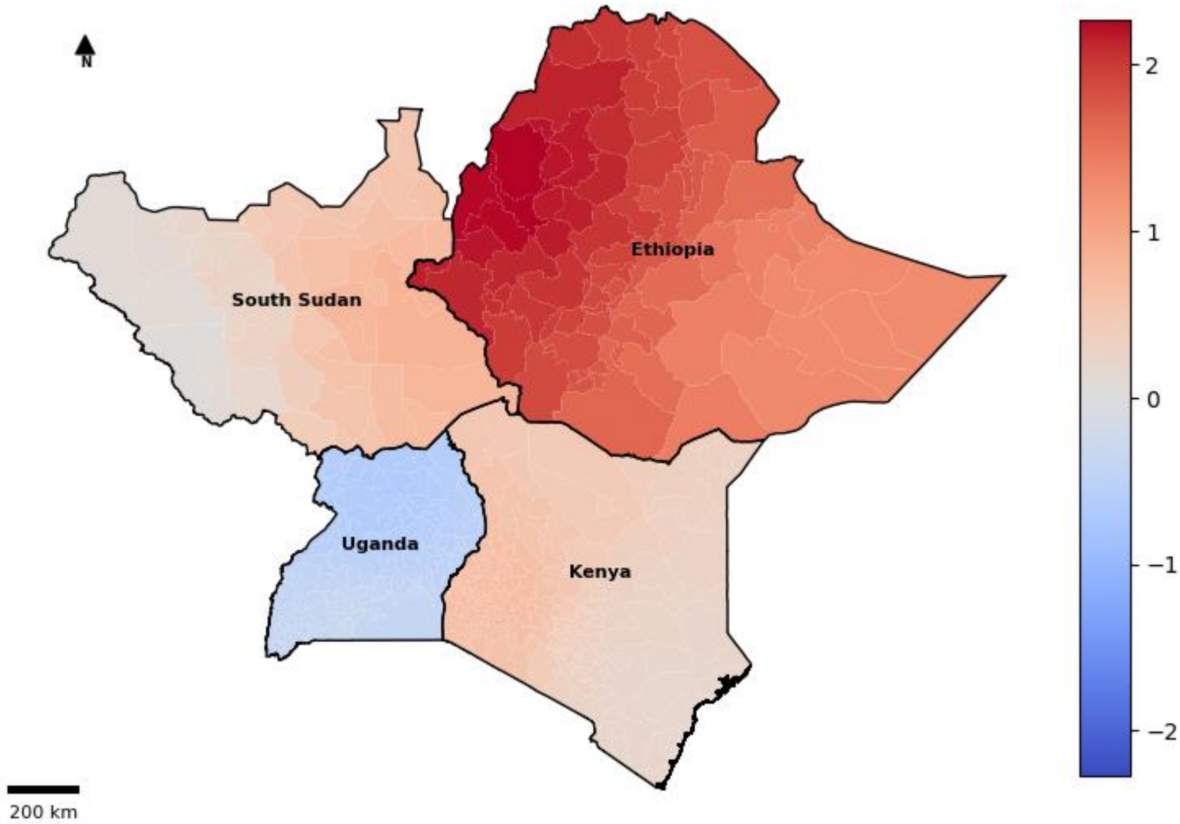

Figure 1. *A map of the police station coefficient for each GWR. Red colouring is associated with a positive correlation with conflict while blue indicated the negative.*

Looking at the plot of Ethiopia, the rate of conflict is more strongly correlated with police in the west than in the east. The rate of conflict is also more strongly correlated with police in Ethiopia than in other countries. South Sudan and Kenya also show slight east-to-west spatial variability with positive correlation throughout. Uganda however shows weak negative correlation, suggesting fewer police stations is associated with higher conflict.

Many of the higher conflict areas in Ethiopia (such as Tigray, Amhara, and Oromia) are in the west and these higher conflict zones have had state security responses, so the identified correlation makes sense. These areas have seen significant police and paramilitary deployment, particularly during large scale conflicts and protests (Yishak Endris, 2025). Rather than representing deterrence, this high correlation may represent the state reactivity to violence instead.

For Kenya and South Sudan, the east-to-west gradient in correlation likely mirrors regional disparities in state capacity. In Kenya, for example, areas near the Turkana-Karamoja border are known for cross-border pastoralist conflicts, where police presence is limited and supplemented by local armed groups or community policing efforts (Mulugeta, 2017). Similarly, in South Sudan, the state's inconsistent policing presence is often focused on areas central to peacekeeping or state-building initiatives rather than conflict prevention (Justin, 2017).

In contrast, Uganda's weak negative correlation suggests that areas with fewer police stations experience more conflict. This may indicate that where policing is more consistent and community-based, as in parts of central and southern Uganda, conflict levels are lower.

These findings suggest that the presence of police infrastructure does not uniformly reduce conflict. In some contexts, like Uganda, police presence may represent preventive capacity, while in others, like Ethiopia, it may signal zones of active state contention.

Figure 2 plots the hospital coefficient, representing access to healthcare's correlation with conflict.

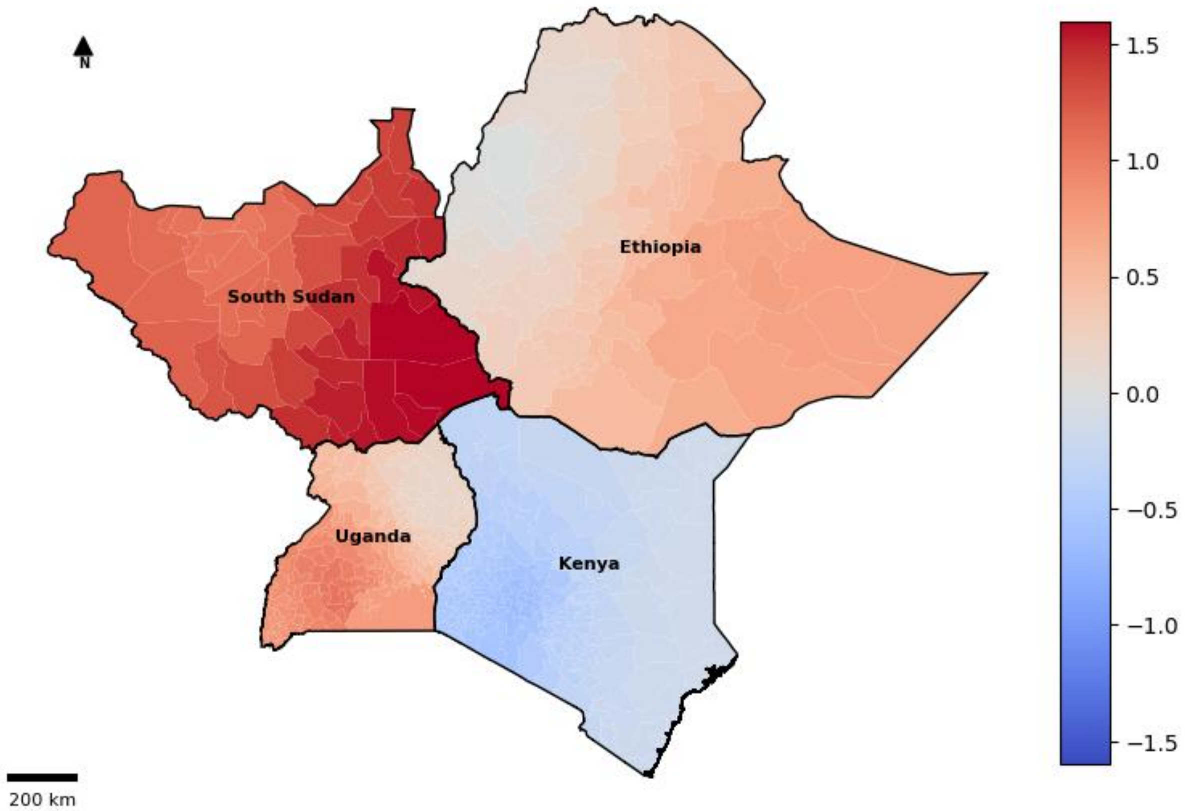

Figure 2. *A map of the hospital's coefficient for each GWR. Red colouring is associated with a positive correlation with conflict while blue indicated the negative.*

This time South Sudan shows the strongest correlation with conflict, strongest in the east with a gradient to the west. Uganda is weakly stronger in the south than the north, and Ethiopia weakly stronger in the east. Kenya has a weak negative correlation, strongest in the central south-west.

These findings suggest healthcare infrastructure interacts with conflict dynamics rather than simply mitigating them. In the case of South Sudan, where the strongest positive correlation was found (strongest in the east tapering to the west). This may indicate that where hospitals are most common, they may be located in zones of active humanitarian intervention or possibly where the state is responding to conflict hotspots. In Africa, healthcare facilities often become emergency hubs in high-violence areas (Nasreldin et al., 2024) so this correlation could be due to the ongoing civil war in South Sudan.

The weak negative correlation in Kenya is interesting as an exception. It may suggest that in Kenya, hospital presence may reflect stronger state legitimacy and security, and therefore a reduction in conflict.

These findings suggest that the access to healthcare does not uniformly correlate to conflict. In some contexts, like South Sudan, hospitals may represent a response to violence and associated injuries, while in others, like Kenya, it may signal areas of higher security.

Figure 3 plots the coefficient for schools.

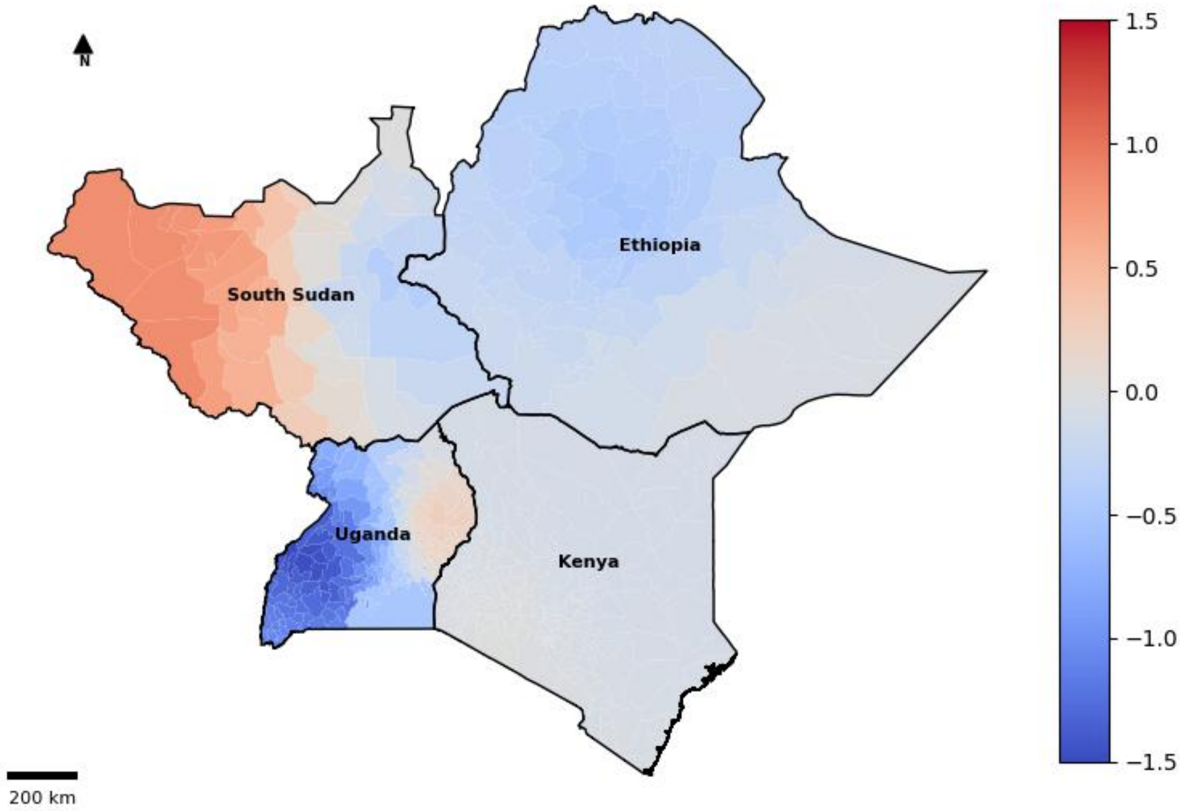

Figure 3. *A map of the school's coefficient for each GWR. Red colouring is associated with a positive correlation with conflict while blue indicated the negative.*

Uganda's schools have a strong negative correlation with conflict, whereas South Sudan's are positively correlated in the west and negatively correlated in the east. Kenya shows very little spatial variability, and Ethiopia has a weak negative correlation centrally.

In Uganda, the strong negative correlation between the number of schools and conflict suggests that areas with greater educational access may correspond with lower conflict risk. This makes sense as literature shows that schools can function as stabilising social institutions and reduce youth grievance formation (Education Cannot Wait, 2024; NRC, 2017).

In South Sudan, the pattern of positive correlation in the west and negative in the east may reflect that in the west, educational infrastructure is in areas affected by conflict where the state or humanitarian groups have attempted to establish presence (IDS, 2024; GCPEA, 2024).

Figure 4 plots the R-squared value for the GWR.

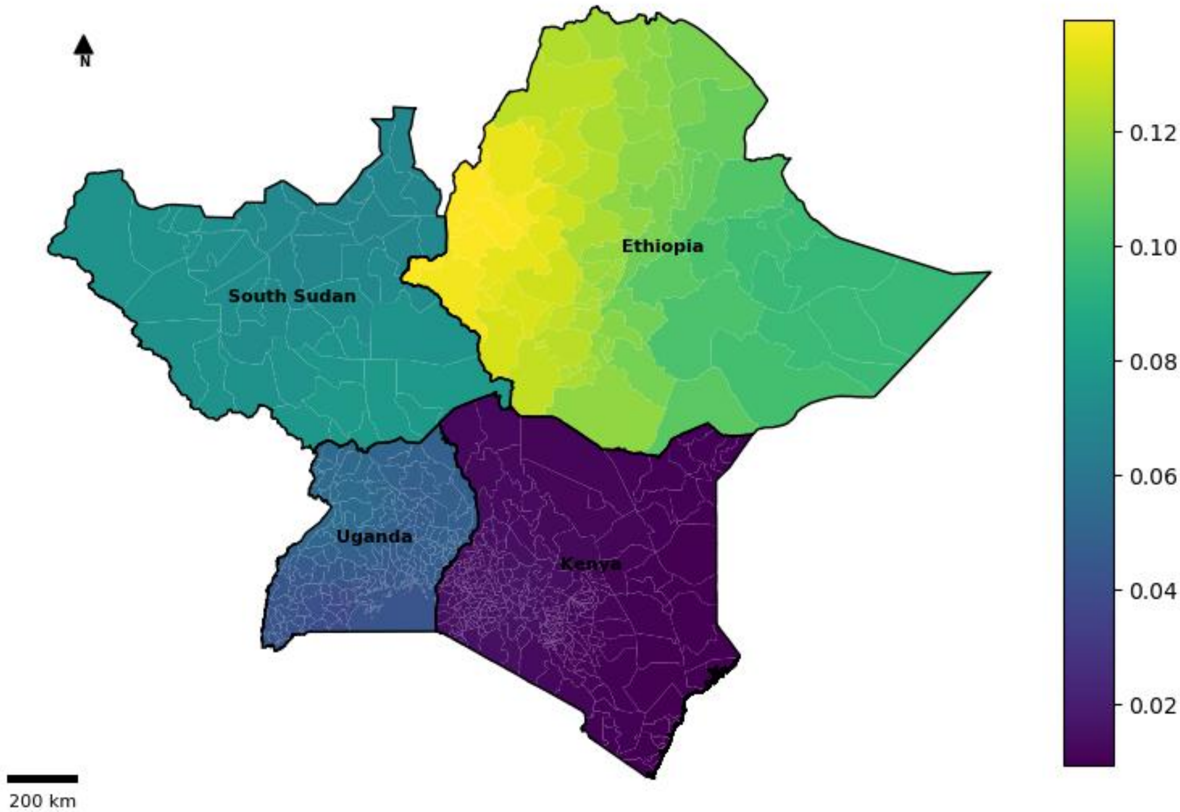

Figure 4. *A map of the R-squared value for the GWR. This value represents the variance explained by the model.*

The r-squared values for Kenya, Uganda, and South Sudan do not appear to greatly vary with space (~0.02, 0.05, 0.08 respectively). Ethiopia has a higher value in the west (~0.13,) and slightly lower in the east (~0.10).

The values for Kenya, Uganda, and South Sudan are low and don't vary, suggesting the predictors don't explain much of the variation in conflict in these countries, and the spatial context doesn't strongly improve the explanatory power of the model.

The values for Ethiopia are still small but suggests that the predictors are somewhat more useful in explaining conflict patterns, especially in the west. The GWR may be more informative in this case.

Table 1 compares the GWR output with the baseline OLS model.

Table 1. *A visual representation of conceptual workflow. This flowchart was made with assistance from generative AI.*

Country	OLS AIC	OLS AICc	GWR AICc	Difference (OLS - GWR)
Kenya	2003.43	2003.56	2005.92	-2.35
Ethiopia	515.25	515.78	518.42	-2.64
South Sudan	321.78	322.78	327.11	-4.32
Uganda	1107.02	1107.26	1107.24	0.02

Lower AICc values indicate a better-fitting model. The table suggests that the GWR performs worse than the OLS in all except Uganda where the models are close to identical. This suggests that the link between predictors and conflict are too weak and the added complexity of the GWR isn't justified.

4. CONCLUSIONS

This analysis explored the spatial relationship between conflict and proxies for state presence, healthcare and education access. While the overall explanatory power of the model was weak, spatial patterns with meaningful implications were identified. Ethiopia showed stronger spatial variation in both correlation strength and R-squared values, particularly in the west. Uganda stood out as a strong negative correlation between schools and conflict, suggesting stabilisation of education access. Conversely in South Sudan positive correlations suggest infrastructure may be concentrated in areas of conflict. Kenya showed very little spatial variation of coefficients.

Interestingly the GWR offered no improvement over OLS, suggesting relationships are weak or inconsistent, and results from this study likely cannot be relied upon for any real use-case.

5. REFERENCES

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